Subject	**DATABASE SYSTEMS**			
	Type	**Semester**	**ECTS**	**Code**
	OBLIGATIVE (O)	3	5	
Course Lecturer	Elton Boshnjaku			
Course Assistants	Alma Novoberdaliu & Rigon Pira			
Aims and Objectives	This course offers students the study of Database Systems, Defining Requirements for the Database systems, including Database Design with Entity Relationship-Based Models, Relational Modelling, and Using SQL and Relational Algebra. The student will power with knowledge at modelling databases from conceptual and physical levels of design, then be able to develop database schemas that enforce data integrity. Including knowledge in creating and manipulating of SQL operation to enforce those operation of database schemas.			
Prerequisite(s)	Recommended: concepts of system requirements, knowledge of defining objects but not necessarily			
Learning Outcomes	Upon completion of the course, the student should be able to: • Understand and understand the functions offered by Database Management Systems, as well as the choice of DBMS for creating real applications. • Knowledge of modelling conceptual and logical level of database design. Use the entity-relationship model (ER& EER) and ER diagrams • Implement a relational database schema using Structured Query Language (SQL), to create and manipulate tables, indexes, and views. • Create and use simple, intermediate and complex queries in SQL.			
12 Weeks Course Content	**Course Plan** **Unit I** - Introduction to Database Concepts **Unit II** - Presentation of Data Models **Unit III** - Introduction to SQL (DDL & DML) **Unit IV** - Conceptual Model (ERD - Entity Relationship Diagram) **Unit V** - Enhanced Conceptual Model (EERD - Enhanced Entity Relationship Diagram) **Unit VI** - Logical Database Model **Unit VII** - Physical Model (DDL - Data Definition Language) **Unit VIII** - Data Manipulation with SQL (DML - Cartesian Product) **Unit IX** - Data Manipulation with SQL (DML - JOINs) **Unit X** - Data Manipulation with SQL (DML - Subqueries) **Unit XI** - Creating Summaries with SQL (WITH, VIEW) and Stored Procedures **Unit XII** - Normalization (1NF, 2NF) and Relational Algebra			
Teaching/Learning Methods	**Teaching/Learning Activity**			**Weight (%)**
	1. Lectures			25%
	2. Laboratory Exercises			25%
	3. Self Study			50%
Assessment Methods	**Assessment Activity**	**Number**	**Week**	**Weight (%)**
	1. Test	2	6,12	20%
	2. Final Exam			70%
	3. Assignments			10%
	4. Project	1	13	??
Course resources	**Resources**			**Number**
	1. Classes			1
	2. Laboratory			1
	3. Moodle			1
	4. Software: Microsoft SQL Server Management Studio, Draw.io			1
	5. Projector			1

	Activity	Weekly hrs	Total workload
ECTS Workload	1. Lecture	2	30
	2. Seminars		
	3. Laboratory	2	30
	4. Practice in industry		
	5. Independent learning	5	88
	6. Final Exam	1	2
Literature/References	• Fundamentals of Database Systems, 7th Edition, 2016, Ramez Elmasri and Shamkant B. Navathe • Practical SQL: A Beginner's Guide to Storytelling with Data, 1st Edition, 2018, Anthony DeBarros • SQL All-in-One For Dummies, 3rd Edition, 2019, Allen G. Taylor		
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